

Nested Loops & Exit Statement

ELEC1006: ENGINEERING COMPUTING

Combining Loops

- Loops can be combined together (nested loops).
 - The statement within a loop can contain another loop:

```
int i, j;  
for (i = 0; i < 4; i++)  
{  
    for (j = 0; j < 4; j++)  
    {  
        // statements  
    }  
}
```

Nested Loops

- Inner loop goes through all repetitions for each repetition of outer loop
- Inner loop repetitions complete sooner than outer loop
- Total number of repetitions for inner loop is the product of number of repetitions of the two loops.

exit() Statement

- Use `exit(n)` to exit the program.
 - `n` is a specific exit code:
 - 0 for successful exit.
 - 1 or non-zero number for failure as mentioned below.
- **exit(1)**: It indicates abnormal termination of a program perhaps as a result a minor problem in the code.
- **exit(2)**: It is similar to `exit(1)` but is displayed when the error occurred is a major one. This statement is rarely seen.
- **exit(127)**: It indicates command not found.
- **exit(132)**: It indicates that a program was aborted (received **SIGILL**), perhaps as a result of illegal instruction or that the binary is probably corrupt.
- **exit(133)**: It indicates that a program was aborted (received **SIGTRAP**), perhaps as a result of dividing an integer by zero.
- **exit(134)**: It indicates that a program was aborted (received **SIGABRT**), perhaps as a result of a failed assertion.
- **exit(136)**: It indicates that a program was aborted (received **SIGFPE**), perhaps as a result of floating point exception or integer overflow.
- **exit(137)**: It indicates that a program took up too much memory.
- **exit(138)**: It indicates that a program was aborted (received **SIGBUS**), perhaps as a result of unaligned memory access.
- **exit(139)**: It indicates **Segmentation Fault** which means that the program was trying to access a memory location not allocated to it. This mostly occurs while using pointers or trying to access an out-of-bounds array index.
- **exit(158/152)**: It indicates that a program was aborted (received **SIGXCPU**), perhaps as a result of CPU time limit exceeded.
- **exit(159/153)**: It indicates that a program was aborted (received **SIGXFSZ**), perhaps as a result of File size limit exceeded.

Example

```
#include <iostream>
using namespace std;

int main ()
{
    unsigned long n=1;
    for (n; n<5; n++)
    {
        cout << "hello world\n";
        if(n==2)
            exit(1);
    }

    cout << "this is the end of the program.";

    return 0;
}
```

More info

- [1] cplusplus.com: Statements and flow control
<https://cplusplus.com/doc/tutorial/control/>
- [2] cplusplus.com: function exit
<https://cplusplus.com/reference/cstdlib/exit/?kw=exit>
- [3] learncpp.com: 5.7 – For statements
<https://www.learncpp.com/cpp-tutorial/57-for-statements/>
- [4] Exit codes in C/C++ with Examples
<https://www.geeksforgeeks.org/exit-codes-in-c-c-with-examples/>